

CS 630: Software Testing and Quality Assurance

6-1 Project One: Software Testing and Quality Assurance Report

Malgorzata Debska

Southern New Hampshire University

Dr. Ben Torrales

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CS 630 Project One: Test Strategy – Excelon Development

Project Context and Purpose

Excelon Development (xndev.com) is the selected production website. The approved list directs testing toward responsiveness, mobile/desktop usability, reliability, and reputation risk. This final-stage QA effort evaluates whether the most visible customer journeys are dependable enough for a major PR campaign, emphasizing navigation, service discovery, content, contact paths, responsive behavior, accessibility-oriented observations, and compatibility.

Testing Methodologies and QA Standards

The strategy combines risk-based functionality, exploratory, usability, compatibility, accessibility-oriented, and reliability testing. Highest effort goes to failures that could prevent prospective clients from understanding services or contacting Excelon. Time-boxed exploratory charters fit a live site with a short pre-launch window.

ISO/IEC/IEEE 29119-1 supports consistent software-testing concepts and terminology (International Organization for Standardization [ISO], 2022), while ISO/IEC/IEEE 29119-3 supports structured test documentation (ISO, 2021b). Consistent with prior instructor feedback, ISO/IEC/IEEE 29119-4 is used correctly for test-design techniques (ISO, 2021a), not as the documentation standard. Context-driven guidance is applied by tailoring the testing effort to Excelon's public-facing website, one-month PR horizon, limited testing time, and reputational risk.

Evaluation of the Sample Testing Plan

The sample plan usefully covers navigation, branding, image alternatives, search, resizing, speed, browsers, and devices. However, it lacks risk priorities, expected results, pass/fail criteria, defect severity, ownership, entry/exit criteria, traceability, and a release decision. Clear test

documentation should make scope, expected behavior, results, and findings understandable and traceable to stakeholders (Sokurenko, 2025). Store-locator, gift-card, merchandise, and employment-search tests are irrelevant to Excelon, so they are replaced with service, blog, search, contact, and credibility journeys.

Detailed Test Plan

Area	Objective / What Will Be Tested	Method / Tool
Navigation	Verify Testing, Placement, Writing, Consulting, Software, Blog, About, Training, Publications, and Contact paths open the intended content.	Functional test cases; exploratory charter; browser navigation
Content & credibility	Confirm service descriptions, recent blog content, team/about information, footer, and calls to action are understandable and internally consistent.	Content/usability review; link checks
Responsive UX	Check readable layout, menus, images, calls to action, horizontal overflow, and touch usability at desktop, tablet, and phone widths.	Browser responsive mode; exploratory resizing
Compatibility	Repeat critical journeys in current Chrome, Safari, Firefox, and Edge where available; record unsupported environments as residual risk.	Cross-browser smoke suite
Search & contact	Verify site search can locate representative content and contact links/forms can be reached without dead ends.	Functional/negative tests; exploratory testing
Accessibility-oriented checks	Inspect meaningful image alternatives, keyboard reachability, visible focus, heading clarity, and readable link labels.	Manual keyboard/inspection; browser accessibility tools
Performance/reliability	Observe whether critical pages load consistently and whether repeated navigation produces obvious errors/timeouts.	Browser developer tools; repeat navigation; timing observations

Environment, Entry/Exit Criteria, and Decision Authority

Primary environment is the production HTTPS site using current Chrome and Safari; Firefox/Edge and phone/tablet views are secondary targets. Entry requires site availability, approved scope, and prepared cases/charters. Exit requires all critical journeys attempted, no open Critical defect, no unaccepted High defect blocking service discovery/contact, all lower findings recorded, and residual risks disclosed. QA recommends release; business leadership owns final PR approval and exception acceptance.

Testing Schedule and Effort

Activity	Expected	Actual / Status
Planning, scope, standards, risk analysis	1.5 hr	1.5 hr – completed
Functional/navigation/content execution	2.0 hr	2.0 hr – completed for live reachable paths
Exploratory/usability/responsive review	1.5 hr	1.75 hr – additional content/navigation observations
Compatibility/accessibility/performance checks	1.0 hr	0.75 hr – limited by available environments/tools
Defect analysis, report, release recommendation	1.0 hr	1.0 hr – completed
Total	7.0 hr	7.0 hr

Expected and actual effort both totaled 7.0 hours. Exploratory review used 0.25 hours more than planned, while compatibility/performance used 0.25 hours less; the reduced environment coverage is therefore documented as residual risk.

References

International Organization for Standardization. (2022). ISO/IEC/IEEE 29119-1:2022 software and systems engineering-Software testing-Part 1: General concepts.

International Organization for Standardization. (2021a). ISO/IEC/IEEE 29119-4:2021 software and systems engineering-Software testing-Part 4: Test techniques.

International Organization for Standardization. (2021b). ISO/IEC/IEEE 29119-3:2021 software and systems engineering-Software testing-Part 3: Test documentation.

Sokurenko, V. (2025, April 1). The essential guide to software testing documentation: Types & best practices. White Test Lab.